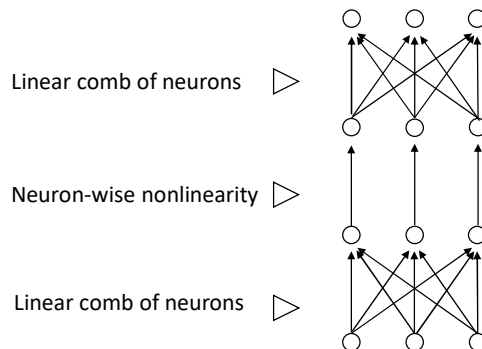


Multilayer Perceptron



- + Universal
- + Simple (elegant theory)
- + Embarrassingly parallel
- Weak inductive biases
- Sample inefficient / data hungry
- Dense (fully-connected) linear layers take a lot of compute

1

Problems with FCN for Images

- Can't cope with inputs of different lengths
 - Images of different sizes (later text)
- Large number of parameters to train
 - Example: 224×224 RGB image = 150,528 dimensions
 - Hidden layers generally larger than inputs
 - One hidden layer = $150,520 \times 150,528$ weights = ~22 billion
- Nearby pixels statistically related
- Should be stable under transformations
 - Don't want to re-learn appearance at different parts of image

We can pin down truth either by adding more data, or by using a more constrained architecture.

2

Networks for images

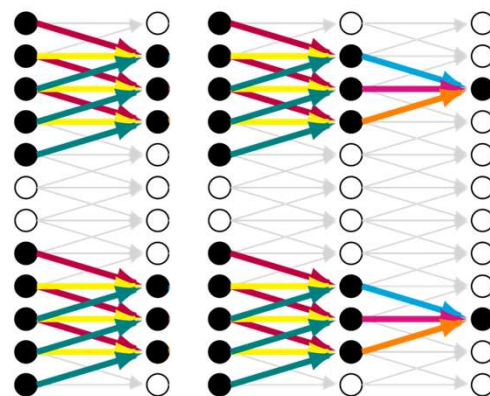
CNN Approach has many useful properties

- Size
 - FCN: One hidden layer = $150,520 \times 150,528$ weights = ~22 billion
 - **CNN**: Parameters are shared
- CNN: Sweep operation
 - Nearby pixels statistically related
 - Should be stable under transformations
 - Don't want to re-learn appearance at different parts of image
 - A way to process variable sized inputs

3

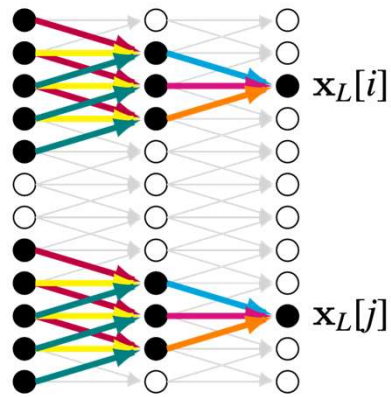
Limitation of CNN

- Processing is local
- Resolved using multiple blocks
 - But attention to "further" parts of the image is delayed while computation is going on
 - The receptive field becomes larger but the neurons don't "talk" to each other while this happens



4

Transformers

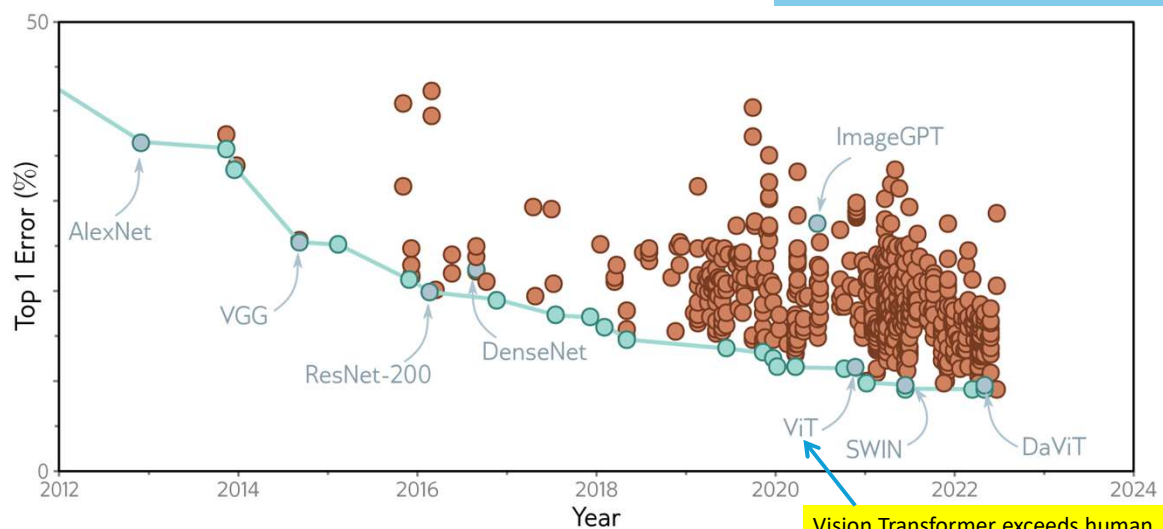


- The receptive field becomes larger but the neurons don't "talk" to each other while this happens
- Attention to other parts of the image is delayed while computation is going on
- A transformer block has CNN-like parameter count but FC-like connectivity, achieved via data-dependent attention

5

Transformers

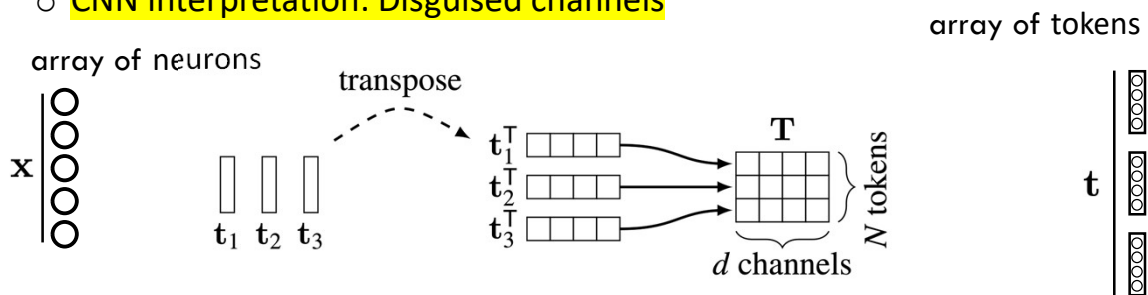
- Three concepts
1. Tokens
 2. Attention
 3. Positional Encoding



6

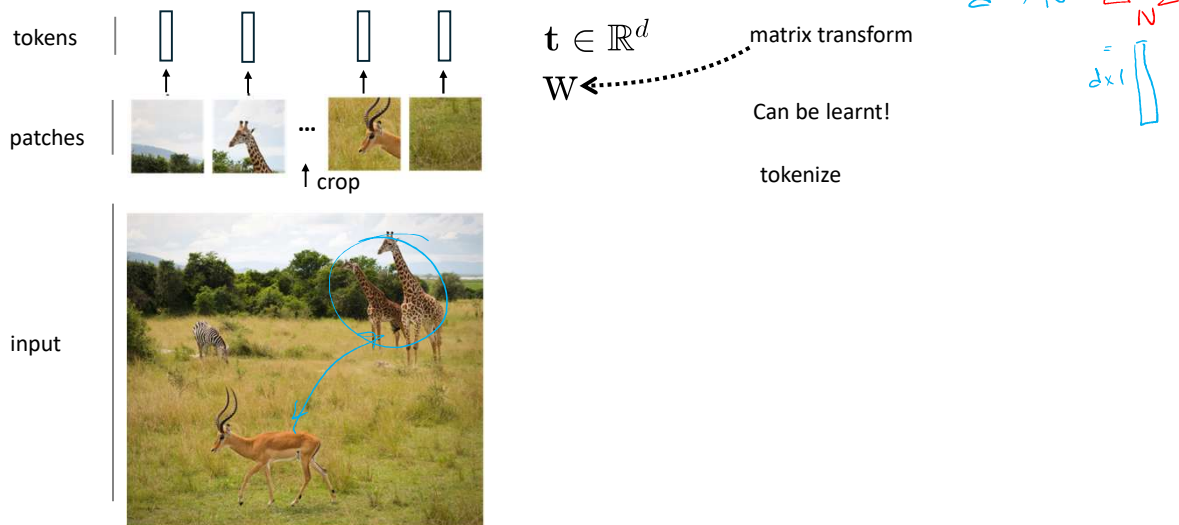
A New Data Type: Tokens

- A token can be thought of as a vector of neurons (“patch”)
 - But the connotation is that a token is an encapsulated bundle of information; with transformers we will operate over tokens rather than over neurons.
- We can also have an array of tokens
- **CNN interpretation: Disguised channels**



7

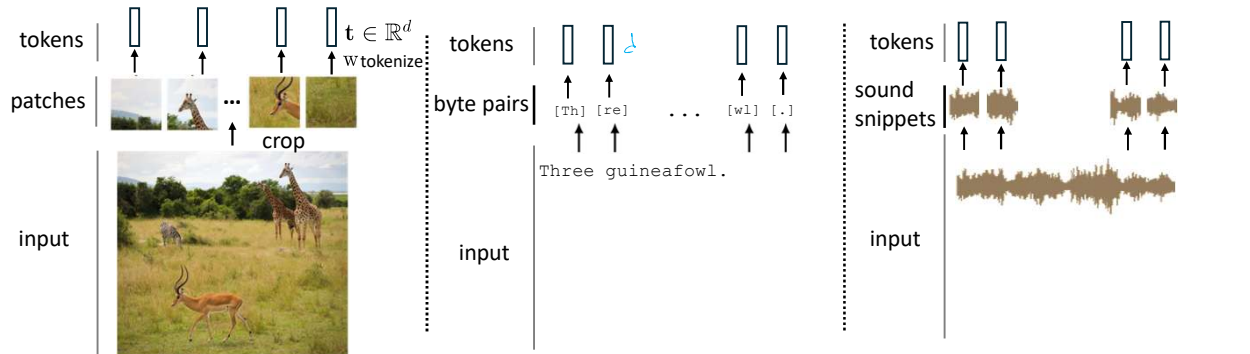
How to get tokens?



8

Tokenizing input data

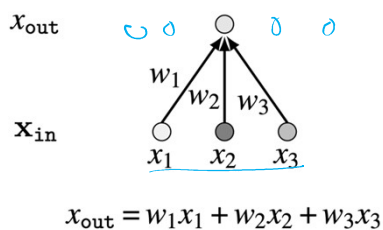
- You can tokenize anything
- General strategy: chop the input into chunks, project each chunk to a vector



9

Operation on Tokens

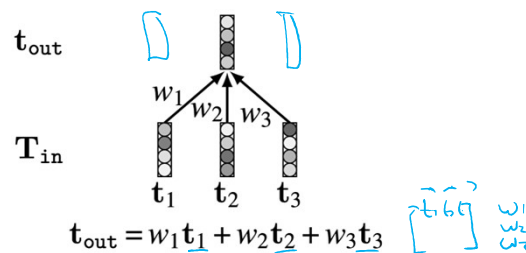
Linear combination of neurons



$$x_{\text{out}}[i] = \sum_{j=1}^N w_{ij} x_{\text{in}}[j]$$

$$\mathbf{x}_{\text{out}} = \mathbf{W} \mathbf{x}_{\text{in}}$$

Linear combination of tokens



$$\mathbf{T}_{\text{out}}[i, :] = \sum_{j=1}^N w_{ij} \mathbf{T}_{\text{in}}[j, :]$$

$$\rightarrow \mathbf{T}_{\text{out}} = \mathbf{W} \mathbf{T}_{\text{in}}$$

10

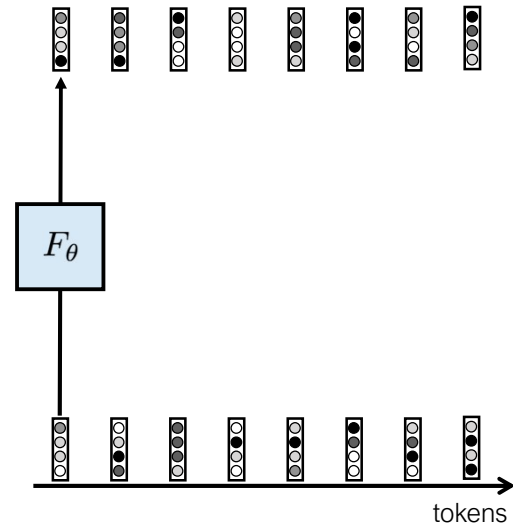
Token-wise nonlinearity

$$\mathbf{x}_{\text{out}} = \begin{bmatrix} \text{relu}(x_{\text{in}}[0]) \\ \vdots \\ \text{relu}(x_{\text{in}}[N-1]) \end{bmatrix}$$

F is typically an MLP

$$\mathbf{T}_{\text{out}} = \begin{bmatrix} F_{\theta}(\mathbf{T}_{\text{in}}[0, :]) \\ \vdots \\ F_{\theta}(\mathbf{T}_{\text{in}}[N-1, :]) \end{bmatrix}$$

The abstractions of the data you want from an ANN is buried in this MLP

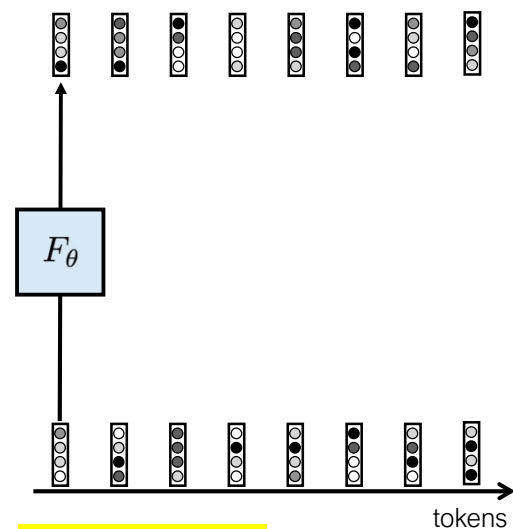


11

Token-wise nonlinearity

$$\mathbf{x}_{\text{out}} = \begin{bmatrix} \text{relu}(x_{\text{in}}[0]) \\ \vdots \\ \text{relu}(x_{\text{in}}[N-1]) \end{bmatrix}$$

$$\mathbf{T}_{\text{out}} = \begin{bmatrix} F_{\theta}(\mathbf{T}_{\text{in}}[0, :]) \\ \vdots \\ F_{\theta}(\mathbf{T}_{\text{in}}[N-1, :]) \end{bmatrix}$$

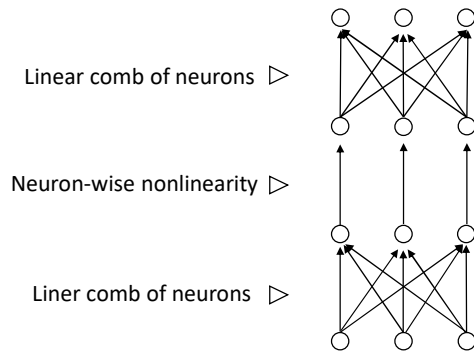


This is like convolution

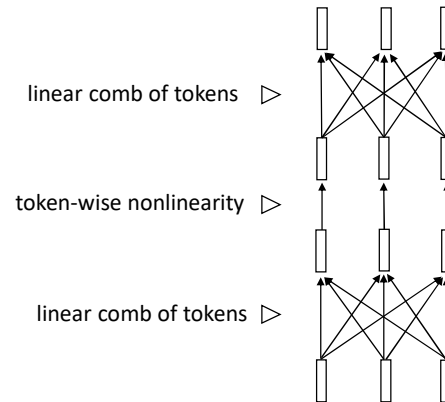
12

Token nets

Neural Net



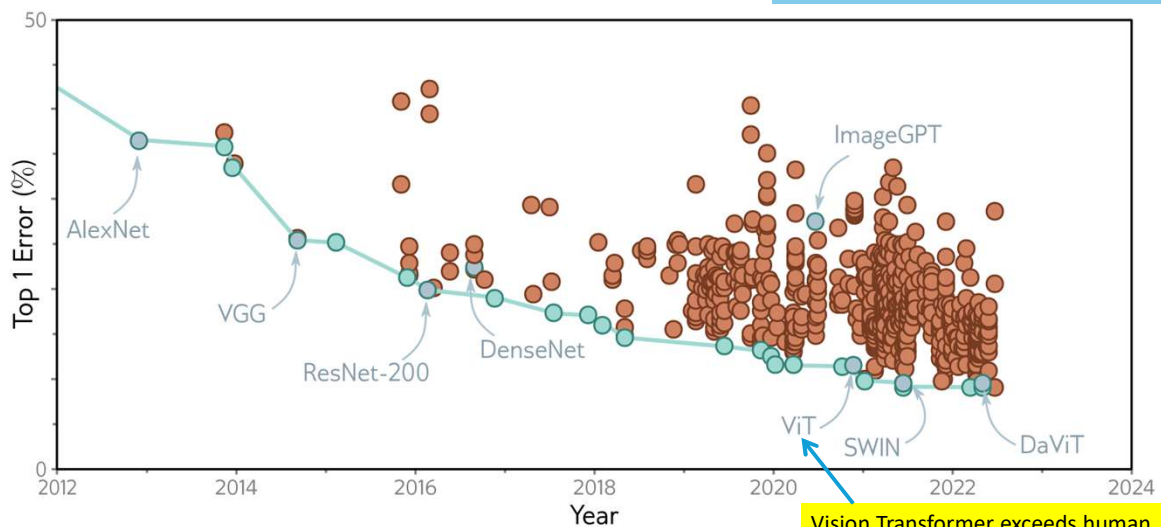
Token Net



13

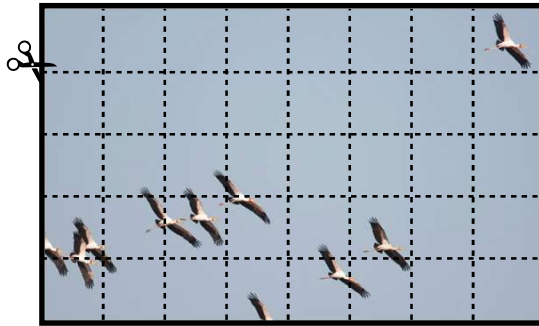
Transformers

- Three concepts
1. Tokens
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 3. Positional Encoding



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Attention is all you need



How many birds are in this image?

Is the top right bird the same species as the bottom left bird?

Where did your gaze travel while you answered this question?

CNNs are built around the idea of locality, and are not well-suited to quickly model long distance relationships

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Relation To NLP

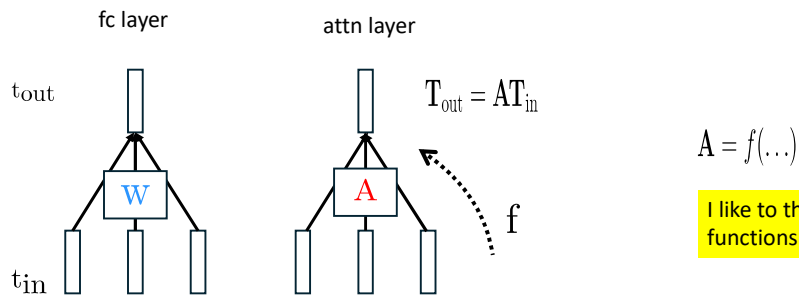
The restaurant refused to serve me a ham sandwich, because it only cooks vegetarian food. In the end, they just gave me two slices of bread. Their ambience was just as good as the food and service.

The word **Their** must “attend to” the word **restaurant**.

- Need to cope with long sentences of variable length
- Sentences contain connections between word representations that depend on the word

16

Parameters vs Activation



I like to think of "sorting" functions in C++

W : free parameters

A is a function of some input data.

A is **attention**: The data tells us which tokens to attend to (assign high weight in weighted sum) to "far away" tokens

17

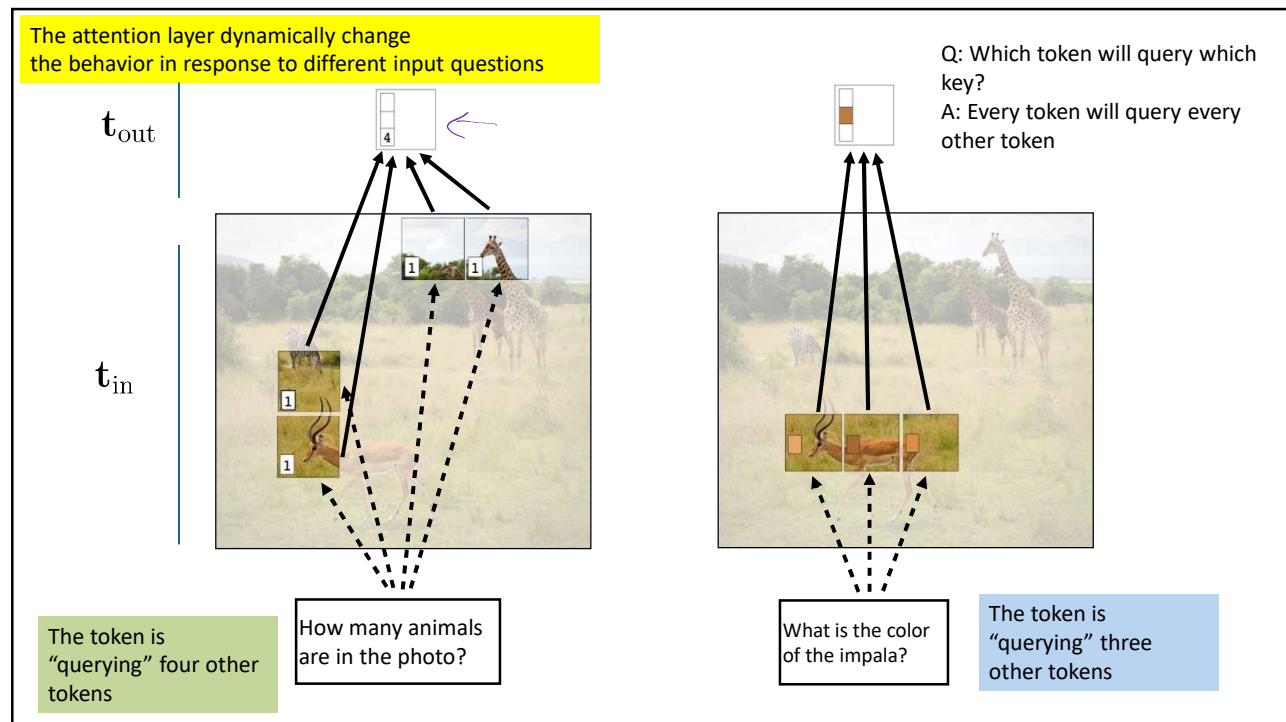
t_{out}

t_{in}

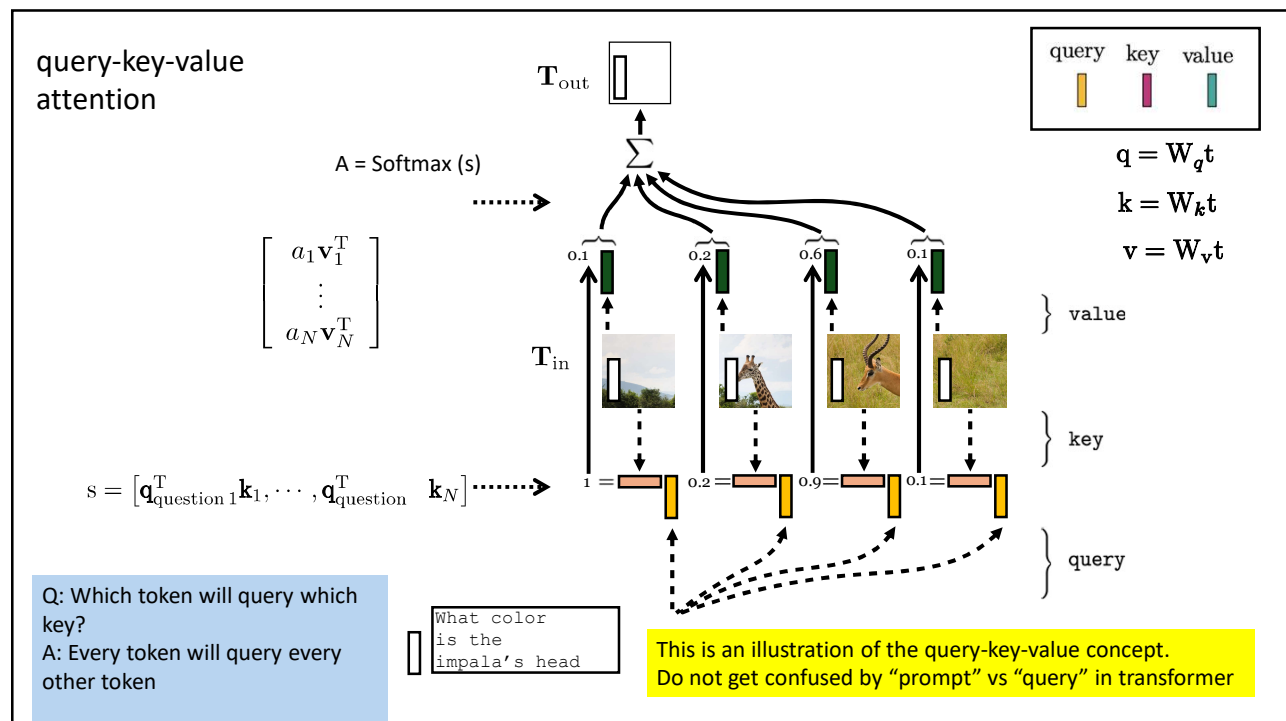


How many animals are in the photo?

18



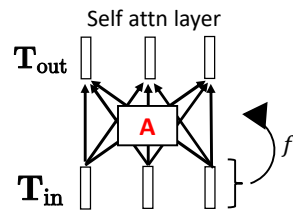
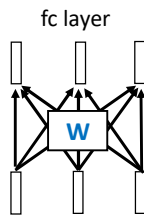
19



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Self-attention

- A classic ANN layer takes an input of size D and outputs a vector of (say) size D using a linear layer followed by a ReLU



A self-attention layer takes N inputs each of size D and returns N tokens

